

Modeling ASPI behavior using Exchange Rate information: Evidence from the Colombo Stock Exchange

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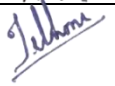
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Declaration

We do hereby declare that the work reported in this report which titled as “Modeling ASPI behavior using Exchange Rate information: Evidence from the Colombo Stock Exchange” is originally prepared by us, Group 08 in the purpose of partial fulfilment of requirement of the Bachelor of Science Degree, Department of Statistics & Computer Science, Faculty of Science, University of Kelaniya, Sri Lanka and not for any other academic purposes. No part of this report has been submitted earlier or concurrently for the same or any other degree.

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Abstract

This study investigates the dynamic relationship between the All Share Price Index (ASPI) and the USD/LKR exchange rate using a combination of time series econometric techniques and machine learning models. The analysis begins with data preprocessing and stationarity testing, where appropriate differencing techniques were applied to remove non-stationarity. The Johansen cointegration test confirmed the existence of a long-run equilibrium relationship between exchange rate movements and stock market performance, and a Vector Error Correction Model (VECM) was used to capture both short-run dynamics and long-run adjustments. In addition, regression techniques including Ordinary Least Squares (OLS), Lasso Regression, and Ridge Regression were implemented to enhance predictive performance and address multicollinearity among lagged variables. Model performance was evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and R^2 . The results indicate the presence of significant short-run interactions and long-run equilibrium behavior between exchange rate fluctuations and ASPI movements, while regularization-based models demonstrate improved forecasting stability, highlighting the effectiveness of combining econometric and machine learning approaches in financial market analysis.

Keywords: *ASPI; USD/LKR Exchange Rate; Cointegration; VECM; Lasso and Ridge Regression*

1.Introduction

1.1 Background of the Study

The stock market is considered a major component of a country's financial system as it reflects economic stability, investor confidence, and overall business performance. In Sri Lanka, the All Share Price Index (ASPI) is widely used to measure the overall performance of the stock market. ASPI represents price movements of all listed companies in the Colombo Stock Exchange and is commonly used by investors and financial analysts to evaluate market trends.

Several macroeconomic factors influence stock market performance, and among them, exchange rate fluctuations play a significant role. The exchange rate between the United States Dollar and the Sri Lankan Rupee (USD-LKR) is particularly important because Sri Lanka is highly dependent on international trade, foreign investments, and external financial transactions. Variations in exchange rates can affect company profitability, import and export activities, inflation levels, and investor behavior.

To analyze this relationship, reliable financial data is essential. In this study, ASPI data was obtained from the Colombo Stock Exchange, while USD-LKR exchange rate data was collected from the Central Bank of Sri Lanka. The study focuses on a one-year period starting from the date when the new President assumed office in Sri Lanka, which represents an important economic transition period in the country.

1.2 Problem Statement

Stock market volatility is influenced by multiple economic factors, making it challenging to accurately predict market movements. Exchange rate fluctuations are believed to have a significant impact on stock market performance, especially in developing economies where external economic conditions strongly influence domestic financial markets. However, the exact nature and strength of the relationship between USD-LKR exchange rate fluctuations and ASPI values remain unclear.

Investors, financial analysts, and policymakers require reliable predictive tools to understand stock market behavior and to minimize financial risks. Without a proper statistical model, it becomes difficult to identify whether exchange rate changes can be used as a reliable predictor of stock market performance. Therefore, it is necessary to develop a statistical model that can analyze and predict ASPI values using USD-LKR exchange rate data.

1.3 Objectives of the Study

Main Objective

The main objective of this study is to develop a statistical model to predict ASPI values using USD-LKR exchange rate data.

Specific Objectives

1. To analyze the relationship between USD-LKR exchange rate fluctuations and ASPI values.
2. To develop and compare multiple predictive models for estimating ASPI values based on exchange rate data, and to identify the most accurate model based on performance evaluation metrics.
3. To assess the accuracy and effectiveness of the developed statistical model.

2. Literature review

Exchange rate fluctuations have been widely recognized as a major macroeconomic factor influencing stock market performance, particularly in developing economies. Exchange rates reflect the relative value of a country's currency and directly affect international trade, foreign investment, and corporate profitability. The All Share Price Index (ASPI) of the Colombo Stock Exchange is commonly used as a key indicator to measure overall stock market performance in Sri Lanka. Understanding the relationship between exchange rate movements and ASPI values is essential for investors, policymakers, and financial analysts to predict market behavior and economic stability.

2.1 Review of Previous Studies

Several empirical studies have investigated the relationship between exchange rate fluctuations and stock market performance using different econometric techniques and datasets.

Study (Anusha, Dhushanthan, & Vinayagathan, 2022) examined the relationship between exchange rate fluctuations and stock market performance in Sri Lanka using the Auto-Regressive Distributed Lag (ARDL) model. The study utilized secondary financial data and analyzed both short-run and long-run relationships between exchange rates and ASPI values. The findings revealed that exchange rates do not have a statistically significant long-run impact on stock market performance. However, the study identified a positive short-run relationship between exchange rate movements and stock market performance. The researchers concluded that exchange rate fluctuations influence investor confidence and foreign investment flows, which indirectly affect stock market dynamics.

Similarly, study (Gooneratne, 2022) investigated the influence of multiple foreign exchange rates, including USD/LKR, GBP/LKR, and EUR/LKR, on stock market performance using long-term financial data obtained from the Central Bank of Sri Lanka and the Colombo Stock Exchange. The study applied the Johansen cointegration technique to identify long-term equilibrium relationships between exchange rate movements and stock market performance. The results indicated that exchange rate fluctuations do not demonstrate a statistically significant long-run relationship with stock market performance. However, the study highlighted that exchange rate volatility contributes to short-term stock market fluctuations and investor decision-making.

In addition to exchange rate-focused research, study (Kengatharan, 2021) analyzed the impact of multiple macroeconomic variables including interest rates, inflation, exchange rates, and money supply on ASPI using an ARDL bound testing approach. The results indicated that exchange rate fluctuations have limited influence on stock market performance in the long run, while interest rates and inflation demonstrated significant negative effects on stock prices. The study emphasized that stock market performance is influenced by the combined effect of several macroeconomic variables rather than a single economic indicator.

Furthermore, study (Balagobei S. &, 2022) examined the relationship between macroeconomic indicators such as GDP growth, exchange rate, interest rate, inflation, and money supply on stock market performance using correlation and regression analysis. The study concluded that exchange rate and interest rate fluctuations negatively affect stock market performance, whereas GDP and money supply have positive effects. The research suggested that macroeconomic stability plays a significant role in maintaining investor confidence and stock market growth.

Earlier empirical evidence presented in study (Menike, 2010) examined macroeconomic influences on stock market returns in emerging financial markets. The study used time-series econometric techniques and found that exchange rate volatility significantly affects stock returns and investment behavior. The research highlighted that currency instability increases financial uncertainty, which can negatively influence stock market stability.

2.2 Methodology Comparison of Previous Studies

Previous studies have employed various econometric techniques to examine the relationship between exchange rates and stock market performance. Studies (Anusha, Dhushanthan, & Vinayagathasan, 2022) and (Kengatharan, 2021) primarily used the ARDL modeling approach to analyze both short-run and long-run relationships between macroeconomic variables and stock market performance. The ARDL approach is widely used because it can effectively analyze time-series data even when variables are integrated at different levels.

Study (Gooneratne, 2022) used the Johansen cointegration technique, which focuses on identifying long-term equilibrium relationships between multiple economic variables. This method is useful in detecting stable long-term associations but may not effectively capture short-term fluctuations in stock market performance.

Study (Balagobei S. &, 2022) mainly used regression and correlation analysis techniques to examine the relationship between macroeconomic variables and stock market performance. Although regression analysis provides useful insights into variable relationships, it may not fully capture dynamic interactions and time-series dependencies.

Study (Menike, 2010) applied time-series econometric techniques in an international context to evaluate the impact of exchange rate volatility on stock market returns. This study emphasized the importance of currency stability in maintaining financial market performance.

2.3 Findings Comparison of Previous Studies

The findings of previous research provide mixed evidence regarding the relationship between exchange rate fluctuations and stock market performance. Studies (Anusha, Dhushanthan, & Vinayagathan, 2022) and (Gooneratne, 2022) reported that exchange rate fluctuations do not significantly influence stock market performance in the long run. However, both studies highlighted the presence of short-run relationships between exchange rate movements and stock market fluctuations.

In contrast, study (Balagobei S. &, 2022) identified a negative relationship between exchange rate fluctuations and stock market performance, suggesting that currency depreciation can reduce investor confidence and corporate profitability. Similarly, study (Menike, 2010) found that exchange rate volatility increases financial risk and negatively affects stock market stability.

Study (Kengatharan, 2021) emphasized that stock market performance is influenced by the combined effect of multiple macroeconomic variables rather than exchange rates alone. These differences in research findings suggest that the relationship between exchange rates and stock market performance may vary depending on economic conditions, data period, and econometric methodology used.

2.4 Research Gap

Although several studies have examined the relationship between exchange rate fluctuations and stock market performance, several research gaps remain. Most previous studies have focused on identifying long-term relationships between macroeconomic variables and stock market performance using broad economic indicators. However, limited research has focused on developing predictive statistical models specifically designed to forecast ASPI values using USD-LKR exchange rate data.

Additionally, many earlier studies used long historical datasets covering extended time periods. These studies may not fully capture recent economic and political transitions in Sri Lanka that could significantly influence exchange rate behavior and stock market performance.

Furthermore, most existing studies primarily focus on correlation and long-run equilibrium relationships rather than short-term predictive analysis. There is limited research focusing on developing practical statistical forecasting models that can be applied for real-world investment decision-making.

Therefore, this study attempts to address these research gaps by developing a statistical model to predict ASPI values using USD-LKR exchange rate data collected during a recent economic transition period. This approach contributes to improving stock market forecasting accuracy and provides practical insights for investors and policymakers.

3.MODEL SPECIFICATION AND METHODOLOGY

3.1 Research Design

This study adopted a quantitative research approach to examine the relationship between the USD–LKR exchange rate and the All Share Price Index (ASPI) in Sri Lanka. The research focused on developing predictive models to forecast ASPI values using historical exchange rate data. Both econometric techniques and machine learning algorithms were utilized to evaluate the predictive strength of exchange rate movements on stock market performance.

The study followed a time series modelling framework which included statistical testing, regression modelling, and machine learning based prediction methods. Multiple model structures were tested to identify the most accurate forecasting model.

3.2 Data Collection

Daily time series data were collected for a period of one year starting from September 2024 to September 2025.

ASPI values were obtained from the Colombo Stock Exchange official website. Exchange rate data representing the USD–LKR conversion rate were obtained from the Central Bank of Sri Lanka database. These sources provide reliable and officially validated financial market information.

3.3 Variables of the Study

Two primary variables were used in this study.

Dependent Variable

- All Share Price Index (ASPI)

Independent Variable

- USD–LKR Exchange Rate

Additionally, lagged variables of both ASPI and USD–LKR exchange rates were introduced to capture delayed market responses and improve forecasting accuracy.

3.4 Exploratory Data Analysis

3.4.1 Time Series Plots

Historical time-series data for the All Share Price Index (ASPI) and the USD-LKR exchange rate were collected to examine the dynamic relationship between stock market performance and currency fluctuations. Time-series plots were generated for both variables to visually compare their movements over the study period. A detailed visual inspection of these plots was then conducted to identify key temporal components, including long-term secular trends and seasonal or cyclic patterns. This preliminary qualitative assessment served as the foundation for characterizing the stochastic behavior of both series prior to formal statistical testing.

3.4.2 Correlation & P- Value

A statistical correlation analysis was performed to examine the interdependence between the ASPI and the USD-LKR exchange rate. The correlation coefficient was calculated to quantify the strength and direction of the linear relationship, while the associated p-values were computed to assess statistical significance and rule out the possibility that the observed association arose from random chance. This approach ensures that the findings reflect a genuine underlying relationship between stock market performance and currency fluctuations.

3.5 Data Pre-Processing

Several data preprocessing techniques were applied prior to model development.

3.5.1 Missing Value Treatment

Missing observations were removed to ensure data consistency and model accuracy.

3.5.2 Descriptive Statistical Analysis

Descriptive statistics were computed to understand the distributional characteristics of the dataset including mean, standard deviation, minimum and maximum values.

3.5.3 Stationarity Testing

The stationarity properties of both time series were formally assessed using the Augmented Dickey-Fuller (ADF) test to ensure the robustness of subsequent econometric analysis and to mitigate the risk of spurious regression. The ADF test results confirmed the presence of unit roots, indicating that the variables were non-stationary in their level form. First-order differencing transformations were subsequently applied to stabilize the mean and variance structures of the series, successfully inducing stationarity and fulfilling the fundamental statistical assumptions necessary for reliable inferential modeling.

3.5.4 Differencing Procedure

The Augmented Dickey-Fuller (ADF) tests showed that both the All Share Price Index (ASPI) and the USD-LKR exchange rate are non-stationary in levels and integrated of order one, $I(1)$. After first-differencing, both series became stationary, $I(0)$, satisfying the stationarity requirement for subsequent time-series analysis.

3.6 Econometric Testing

3.6.1 Granger Causality Test

The Granger causality test was conducted to examine the directional relationship between the USD/LKR exchange rate and ASPI by determining whether past values of one variable can be used to predict the other. The test was performed using stationary data obtained after differencing to ensure reliable results. The null hypothesis states that one variable does not Granger-cause the other, and it is rejected when the p-value is less than the 0.05 significance level, indicating the presence of a causal relationship.

3.6.2 Cointegration Analysis

The Johansen cointegration test was applied to investigate the presence of a long-run equilibrium relationship between the USD/LKR exchange rate and the All Share Price Index (ASPI).

3.6.3 Vector Error Correction Model (VECM)

A Vector Error Correction Model (VECM) was employed to examine the short-run dynamics and long-run equilibrium adjustment mechanisms between the USD/LKR exchange rate and the All Share Price Index (ASPI).

3.7 Model Development

3.7.1 Simple Linear Regression Model

$$ASPI = -137400 + 516.7532(USD_LKR)$$

3.7.2 Multiple Linear Regression model with One lag of USD

$$ASPI = -142300 + 978.3121(USD_LKR) - 444.8750(USD_LKR_lag1)$$

3.7.3 Multiple Linear Regression model with Two lags of USD

$$ASPI = -147000 + 893.6621(USD_LKR) - 68.5217(USD_LKR_lag1) - 275.9044(USD_LKR_lag2)$$

3.7.4 Multiple Linear Regression model with Three lags of USD

$$ASPI = -151200 + 861.2161(USD_LKR) - 112.5408(USD_LKR_lag1) + 89.9212(USD_LKR_lag2) - 275.2154(USD_LKR_lag3)$$

3.7.5 Multiple Linear Regression model with four lags of USD

$$ASPI = -153600 + 708.44(USD_LKR) + 38.18(USD_LKR_lag1) + 21.23(USD_LKR_lag2) + 73.72(USD_LKR_lag3) - 269.72(USD_LKR_lag4)$$

3.7.6 Multiple Linear Regression model with five lags of USD

$$ASPI = -155300 + 599.1744(USD_LKR) + 30.4767(USD_LKR_lag1) + 135.8713(USD_LKR_lag2) + 12.1212(USD_LKR_lag3) + 41.8655(USD_LKR_lag4) - 241.8435(USD_LKR_lag5)$$

3.7.7 Multiple Linear Regression model with five lags of USD and one lag of ASPI

$$ASPI = -1916.5243 - 4.9876(USD_LKR) + 18.0669(USD_LKR_lag1) + 19.5960(USD_LKR_lag2) - 30.1303(USD_LKR_lag3) + 46.4022(USD_LKR_lag4) - 41.6866(USD_LKR_lag5) + 0.9877(ASPI_lag1)$$

3.7.8 Multiple Linear Regression model with five lags of USD and two lag of ASPI

$$ASPI = -1964.8276 - 3.3133(USD_LKR) + 16.6396(USD_LKR_lag1) + 19.4791(USD_LKR_lag2) - 30.7007(USD_LKR_lag3) + 46.9233(USD_LKR_lag4) - 41.5920(USD_LKR_lag5) + 1.0174(ASPI_lag1) - 0.0300(ASPI_lag2)$$

3.7.9 Dynamic Multiple Linear Regression

$$ASPI = -1766.0753 - 1.3112(USD_LKR) + 17.0617(USD_LKR_lag1) - 9.1557(USD_LKR_lag2) + 0.9906(ASPI_lag1)$$

3.7.10 Multiple Linear Regression model with ASPI lag 1

$$ASPI = -1770.0208 + 6.6048(USD_LKR) + 0.9906(ASPI_lag1)$$

3.7.11 Multiple linear Regression model with USD lag 1 and ASPI lag 1(With actual USD value)

$$ASPI = -1789.1577 + 0.0163(USD_LKR) + 6.6487(USD_LKR_lag1) + 0.9907(ASPI_lag1)$$

3.7.12 Multiple Linear Regression model with USD lag 1 and ASPI Lag 1(Without actual USD value)

$$ASPI = -2118.7442 + 7.8683(USD_LKR_lag1) + 0.9890(ASPI_lag1)$$

3.8 Machine Learning Models

3.8.1 Random Forest Model

The Random Forest Regression model was developed using 300 decision trees with a maximum depth of 5 and a minimum of 5 samples required at each leaf node. The model uses two predictor variables: lagged ASPI and lagged USD/LKR exchange rate. The feature importance results indicate that ASPI_lag1 is the most influential variable, contributing approximately 99.12% to the model's predictions, while USD_LKR_lag1 has a very small contribution of about 0.88%. This suggests that the previous period's ASPI value is the primary determinant of the current ASPI level, highlighting strong persistence in stock market movements. The average tree depth of 5 shows that the model maintains moderate complexity, which helps capture nonlinear relationships while reducing the risk of overfitting. Overall, the model demonstrates that historical ASPI values play a dominant role in predicting current market performance, whereas the exchange rate has only a minor additional influence.

3.8.2 Ridge Regression Model

$$ASPI_t = 15346.6414 + (32.2888 \times USD_LKR_lag1) + (1836.4902 \times ASPI_lag1)$$

3.8.3 Lasso Regression Model

$$ASPI_t = 15346.6414 + (24.9855 \times USD_LKR_lag1) + (1850.0672 \times ASPI_lag1)$$

3.9 Model Performance Evaluation

3.9.1 Evaluation Metrics

Mean Squared Error (MSE)

Mean Squared Error (MSE) is a commonly used metric to evaluate the performance of regression models. It measures the average squared difference between the actual observed values and the predicted values generated by the model. A lower MSE value indicates better model accuracy and smaller prediction errors.

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Where:

- Y_i = Actual value
- $\hat{Y}_i$ = Predicted value
- n = Number of observations

Root Mean Squared Error (RMSE)

Root Mean Squared Error (RMSE) is the square root of MSE and represents the prediction error in the same units as the dependent variable. RMSE is useful for interpreting model error magnitude in a more understandable scale. Lower RMSE values indicate better model performance.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}$$

Where:

- Y_i = Actual value
- $\hat{Y}_i$ = Predicted value
- n = Number of observations

Mean Absolute Percentage Error (MAPE)

Mean Absolute Percentage Error (MAPE) measures prediction accuracy as a percentage. It calculates the average absolute percentage difference between actual and predicted values. Smaller MAPE values indicate higher forecasting accuracy.

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{Y_i - \hat{Y}_i}{Y_i} \right|$$

Where:

- Y_i = Actual value
- $\hat{Y}_i$ = Predicted value
- n = Number of observations

R-squared (R^2)

R-squared (R^2), also known as the coefficient of determination, measures the proportion of variation in the dependent variable explained by the independent variables in the model. The R^2 value ranges between 0 and 1, where values closer to 1 indicate better explanatory power and model fit.

$$R^2 = 1 - \frac{\sum (Y_i - \hat{Y}_i)^2}{\sum (Y_i - \bar{Y})^2}$$

Where:

- Y_i = Actual value
- $\hat{Y}_i$ = Predicted value
- $\bar{Y}$ = Mean of actual values

Normalized Mean Squared Error (NMSE)

Normalized Mean Squared Error (NMSE) measures the Mean Squared Error relative to the scale of the actual data, allowing comparisons between models with different magnitudes. A lower NMSE value indicates better predictive accuracy, with values closer to zero representing a better model fit..

$$NMSE = \frac{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{\sigma_Y^2}$$

Mean Absolute Error (MAE)

measures the average absolute difference between the actual values and the predicted values of a regression model. A lower MAE indicates better predictive accuracy, as it reflects smaller average prediction errors in the same units as the original data.

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i|$$

3.9.2 Model Comparison Approach

In order to identify the most suitable model for predicting the All Share Price Index (ASPI), a systematic model comparison approach was adopted in this study. Several statistical and machine learning models, including Linear Regression, Multiple Linear Regression, Random Forest, Ridge Regression, and Lasso Regression, were developed using the same dataset and preprocessing techniques to ensure consistency and fairness in comparison.

The performance of each model was evaluated using multiple accuracy metrics, namely Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R^2). These evaluation measures were selected because they provide comprehensive insights into prediction accuracy, error magnitude, and model explanatory power.

The comparison process was conducted by analyzing the error values produced by each model. Models with lower MSE, RMSE, MAE, and MAPE values indicate better predictive accuracy, as they represent smaller deviations between actual and predicted ASPI values. In contrast, a higher R^2 value indicates a stronger ability of the model to explain the variation in the dependent variable.

All models were trained and tested using identical data splitting and preprocessing procedures to maintain methodological consistency. The final model selection was based on the overall performance across all evaluation metrics rather than relying on a single accuracy measure. This multi-criteria comparison approach ensured that the selected model demonstrates both statistical reliability and practical forecasting accuracy.

The results of this comparison are presented and discussed in detail in the Results and Discussion chapter, where the best-performing model is identified based on the evaluation outcomes.

4.Results and Discussion

4.1 Introduction

This chapter presents the empirical findings of the study and provides a detailed discussion of the results obtained from statistical and econometric analyses. The analysis was conducted to examine the relationship between the USD-LKR exchange rate and the All Share Price Index (ASPI) in Sri Lanka. The chapter includes descriptive statistics, stationarity testing, Granger causality testing, cointegration analysis, Vector Error Correction Model (VECM) results, regression model results, machine learning model results, and model comparison.

4.2 Descriptive Statistics Analysis

Descriptive statistics were calculated to summarize the basic characteristics of the ASPI and USD-LKR exchange rate data. These statistics provide an initial understanding of the data distribution and variability of the variables.

	USD_LKR	ASPI
count	241.000000	241.000000
mean	297.443680	16303.482241
std	3.707825	2615.951987
min	290.259600	11125.470000
25%	294.596300	14654.260000
50%	298.469100	16176.100000
75%	300.670500	17740.460000
max	304.884700	21282.800000

Table 4.2 presents the summary statistics of the variables, including mean, standard deviation, minimum, and maximum values.

The results indicate that ASPI values demonstrated considerable variation throughout the study period, reflecting fluctuations in stock market performance. Similarly, the USD-LKR exchange rate showed notable variability, highlighting exchange rate instability during the selected period.

The standard deviation values suggest that ASPI experienced relatively higher volatility compared to the exchange rate. The minimum and maximum values further confirm the presence of significant fluctuations in both variables during the study period.

The descriptive statistics provide preliminary evidence that the variables exhibit dynamic behaviour, justifying the need for further time series analysis.

4.3 Time Series Visualization

To visually examine the behavior of the variables over the study period, time series plots were generated for ASPI and the USD/LKR exchange rate. Additionally, a combined dual-axis plot was created to facilitate comparative analysis.

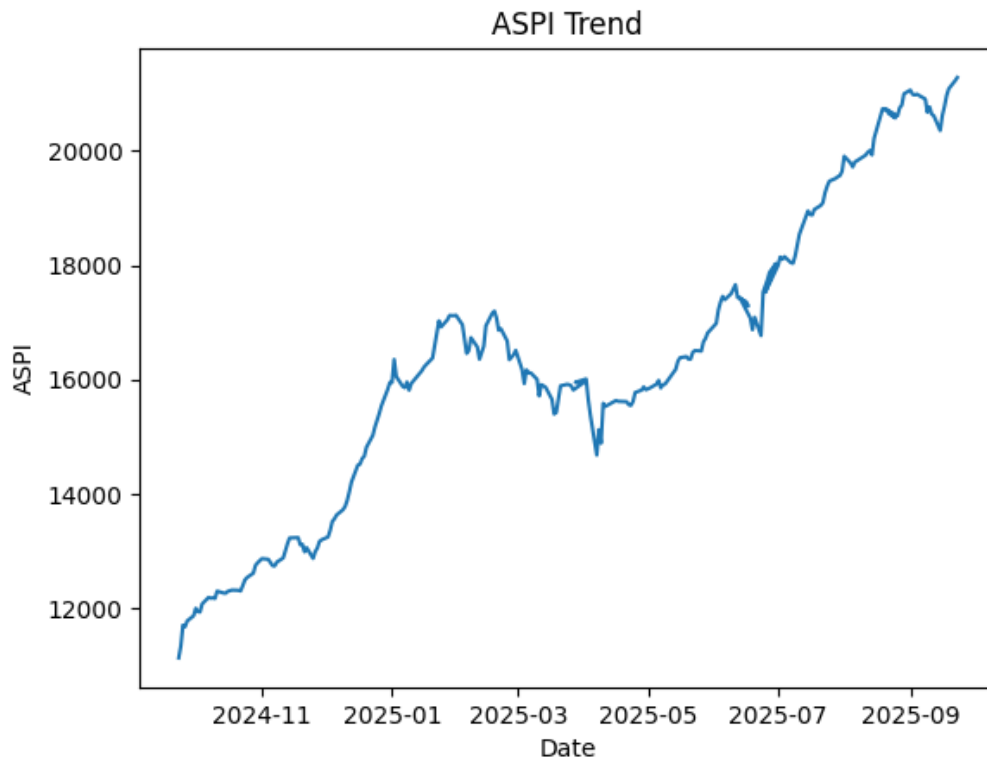


Figure 4.3.1 – ASPI Trend

The ASPI demonstrates noticeable volatility with several short-term fluctuations. However, the index shows an overall upward trend during the study period, indicating improvement in stock market performance and investor confidence. The trend pattern suggests the presence of non-stationarity.

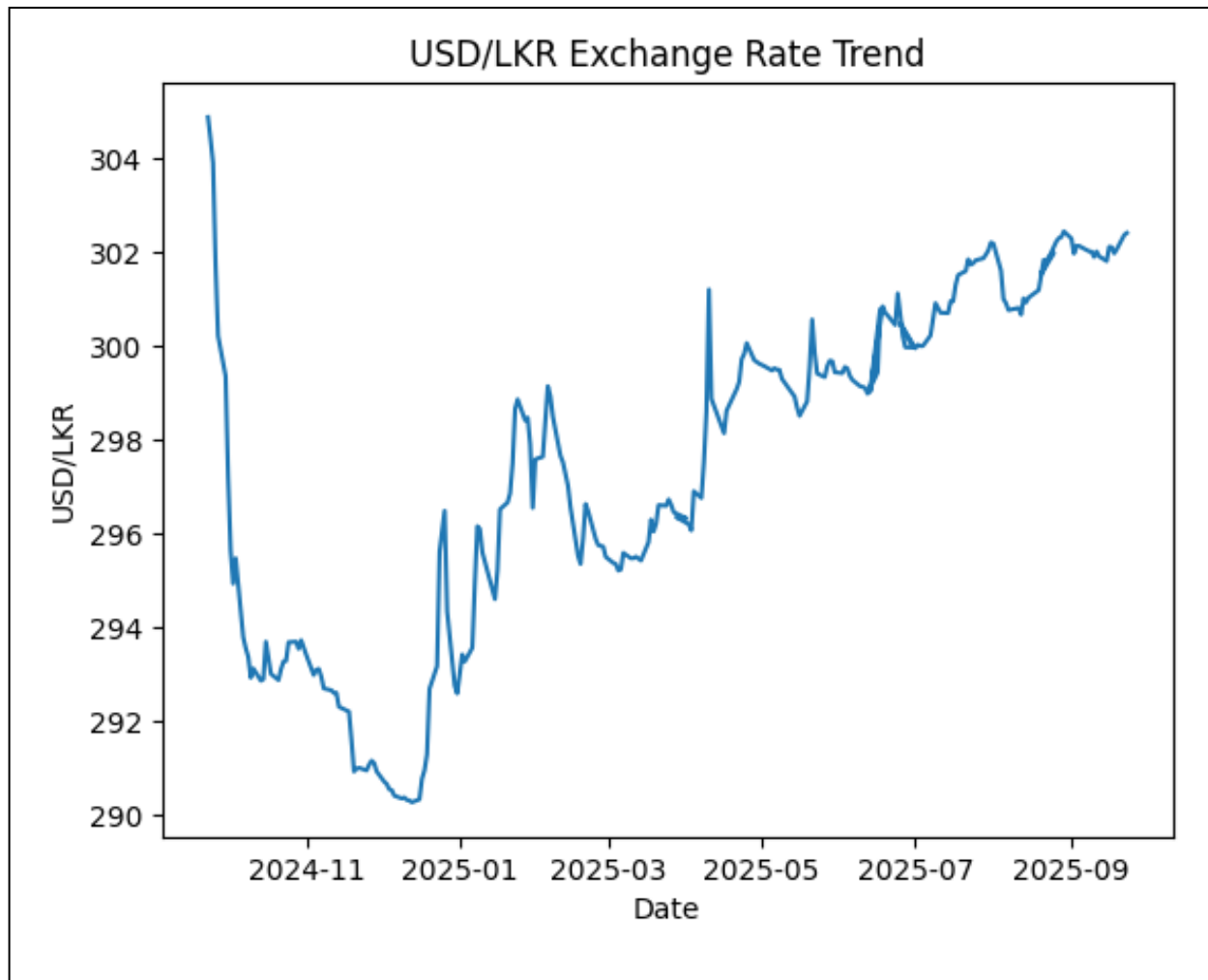


Figure 4.3.2: USD/LKR Exchange Rate Trend

The USD/LKR exchange rate shows short-term fluctuations during the study period. After an initial decline, the exchange rate gradually increased, indicating a depreciation of the Sri Lankan Rupee against the US Dollar. The visible trend suggests possible non-stationary behavior.

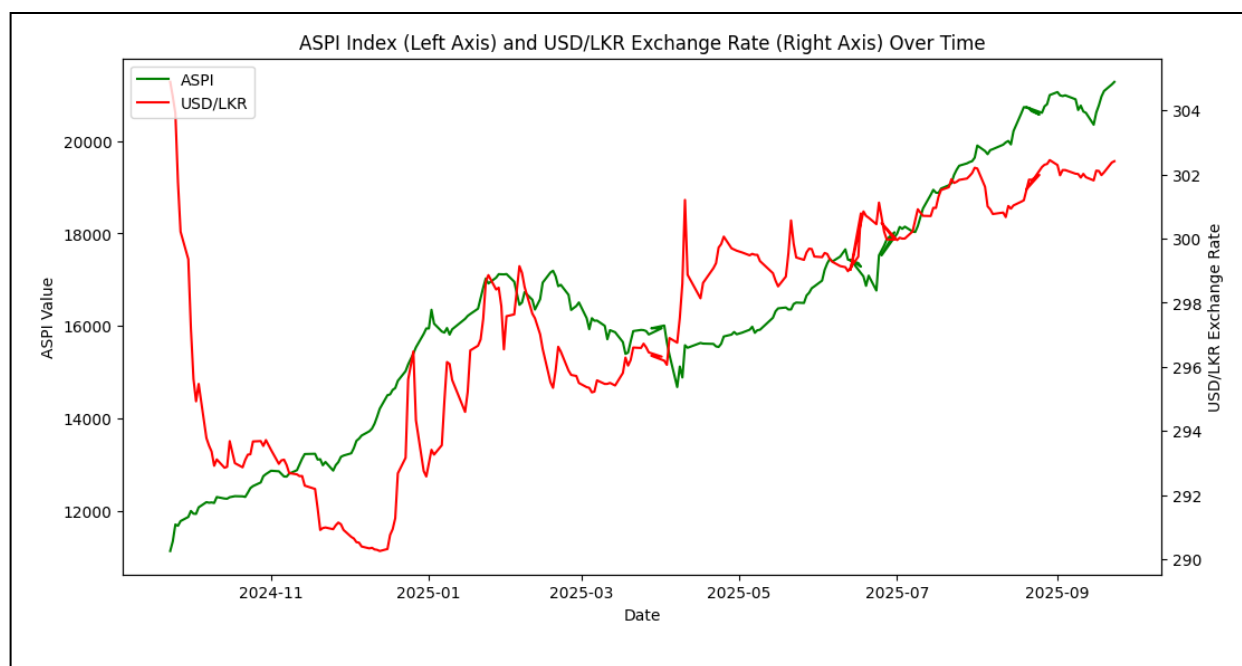


Figure 4.3.3: Combined ASPI and USD/LKR Trend

The combined graph indicates that movements in the USD/LKR exchange rate and ASPI show some level of co-movement over time. Although visual observation does not confirm statistical relationships, the pattern suggests a possible association between exchange rate changes and stock market performance.

4.4 Correlation & P- Value

Pearson correlation coefficient:	0.7324
P-value:	8.76×10^{-42}
Relationship Strength	Strong Positive Relationship
Significance	Statistically Significant

Table 4.4: Correlation & P- Value Results

The Pearson correlation coefficient value of 0.732 indicates a strong positive relationship between the USD/LKR exchange rate and ASPI. The p-value is significantly lower than the 0.05 significance level, confirming that the relationship between the two variables is statistically significant. This suggests that exchange rate movements may have a notable association with stock market performance during the study period.

4.5 Stationarity Test Results

Variable	P-value	Significance Level (0.05)	Stationarity Status	Decision
USD/LKR Exchange Rate	0.5827	0.05	Non-Stationary	Differencing Required
ASPI	0.8573	0.05	Non-Stationary	Differencing Required

Table 4.5: Stationarity Test Results

Stationarity testing was conducted to determine whether the time series variables exhibit constant statistical properties over time. The Augmented Dickey-Fuller (ADF) test was applied to examine the presence of unit roots in both the USD/LKR exchange rate and ASPI series.

Table 4.5 presents the results of the stationarity test for the variables at their level form. The results indicate that the USD/LKR exchange rate recorded a p-value of 0.5827, which is greater than the significance level of 0.05. Similarly, ASPI recorded a p-value of 0.8573, which also exceeds the 0.05 significance level. These results indicate that both variables are non-stationary at their level form, suggesting the presence of unit roots.

Since non-stationary data can lead to unreliable and spurious regression results, first differencing was applied to both variables to achieve stationarity. The transformation ensures that the mean and variance of the series remain stable over time, thereby improving the reliability of subsequent econometric analysis.

The confirmation of non-stationarity at level form justifies the application of differencing techniques and supports the use of cointegration and Vector Error Correction Model (VECM) analysis in the subsequent sections.

4.5.1 Differencing Results

Variable	P-value	Significance Level (0.05)	Stationarity Status	Decision
USD/LKR (First Difference)	0.0000	0.05	Stationary	Suitable for Modelling
ASPI (First Difference)	0.0000	0.05	Stationary	Suitable for Modelling

Table 4.5.1: Differencing Results

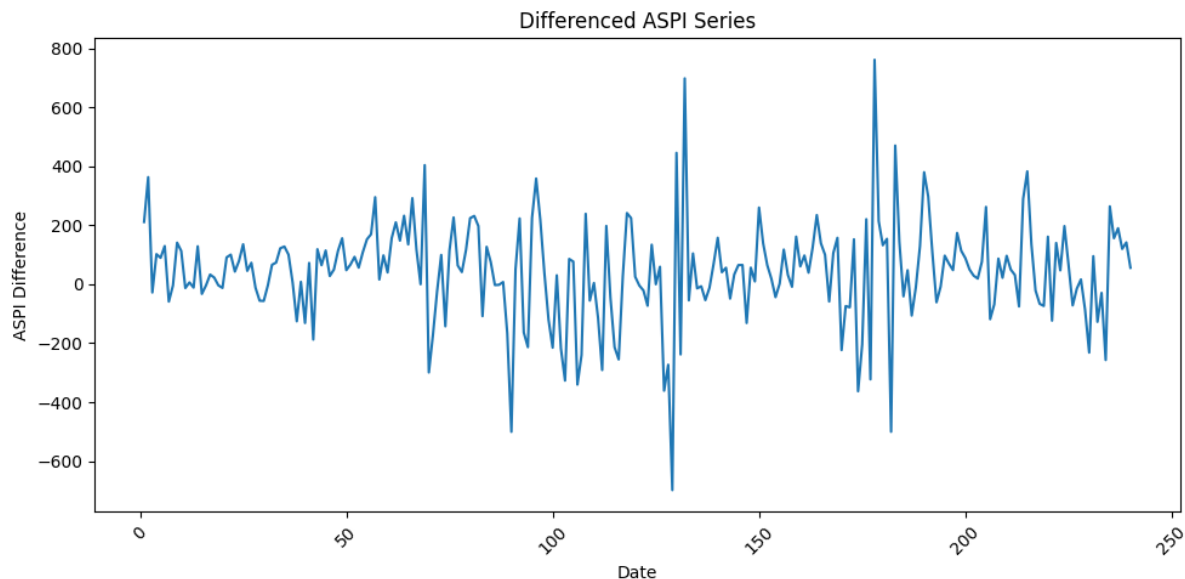
After applying first differencing, the stationarity of both variables was re-evaluated using the Augmented Dickey-Fuller (ADF) test. The results are presented in *Table 4.5.1: Differencing Results*

The first differenced USD/LKR exchange rate recorded a p-value of 0.0000, which is lower than the significance level of 0.05. Similarly, the first differenced ASPI series also recorded a p-value of 0.0000. These results confirm that both variables became stationary after applying first differencing.

The findings indicate that both ASPI and the USD/LKR exchange rate are integrated of order one, $I(1)$. This implies that the variables become stationary only after differencing once. The stationarity of the differenced series ensures that the data is suitable for reliable econometric modelling and prevents the occurrence of spurious regression results.

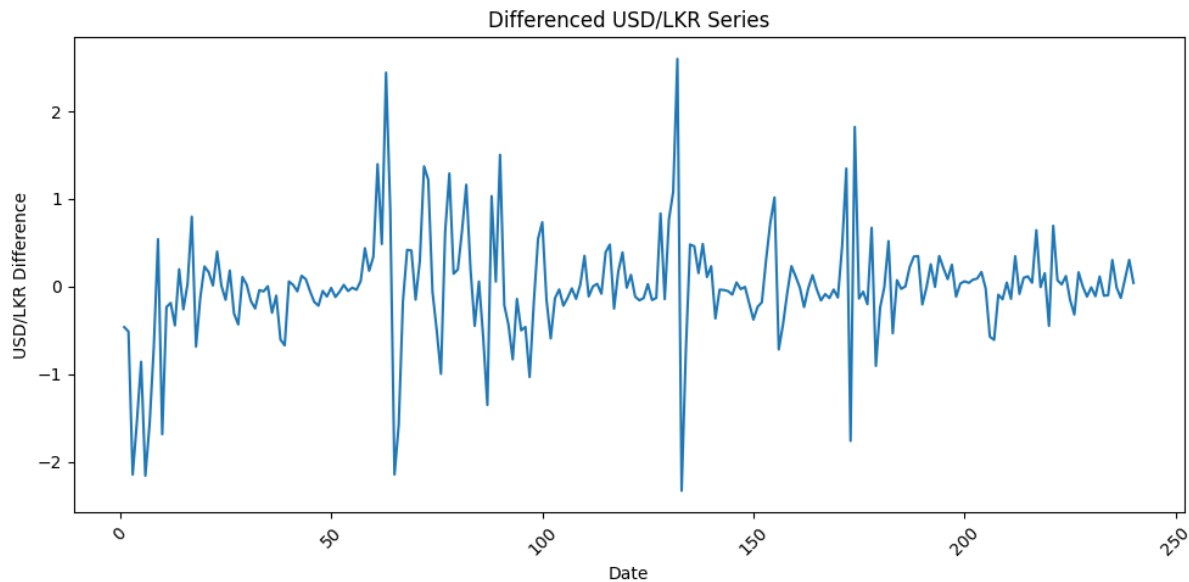
Furthermore, the confirmation that both variables are integrated at the same order supports the use of cointegration testing to examine the existence of a long-run equilibrium relationship between the variables. Therefore, cointegration analysis was conducted in the next section.

Figure 4.5.2: Differenced ASPI Series



The differenced ASPI series fluctuates around a constant mean level with no visible long-term trend. The fluctuations appear more stable compared to the original ASPI series, indicating that differencing successfully removed trend components. This behavior confirms the stationarity of the ASPI series after first differencing.

Figure 4.5.3: Differenced USD/LKR Exchange Rate Series



The differenced USD/LKR exchange rate series also shows random fluctuations around a stable mean value with reduced trend behavior. The absence of a consistent upward or downward movement indicates that the exchange rate series achieved stationarity after differencing.

The graphical behavior of both differenced series supports the ADF test results, confirming that the variables became stationary after first differencing. Therefore, the transformed data was considered suitable for further econometric analysis, including cointegration testing and VECM modelling.

4.5.2 Granger Causality Test Results

The Granger causality test was conducted to examine the directional relationship between ASPI and the USD/LKR exchange rate using stationary data. The test was performed using lag lengths from 1 to 5 to identify whether past values of one variable help predict the other.

Table 4.5.4: Granger Causality Test Results ($ASPI \rightarrow USD/LKR$)

Lag	F-Statistic	P-value	Decision (5%)	Interpretation
1	8.3783	0.0042	Reject H_0	Significant
2	4.2313	0.0157	Reject H_0	Significant
3	2.5328	0.0578	Fail to Reject H_0	Not Significant
4	1.9948	0.0962	Fail to Reject H_0	Not Significant
5	1.4954	0.1923	Fail to Reject H_0	Not Significant

The results indicate that ASPI Granger-causes the USD/LKR exchange rate at lag lengths 1 and 2, as the p-values are less than the 0.05 significance level. However, for higher lag lengths, the relationship becomes statistically insignificant.

Table 4.5.5: Granger Causality Test Results (USD/LKR → ASPI)

Lag	F-Statistic	P-value	Decision (5%)	Interpretation
1	0.0773	0.7812	Fail to Reject H_0	Not Significant
2	1.0679	0.3454	Fail to Reject H_0	Not Significant
3	0.5198	0.6691	Fail to Reject H_0	Not Significant
4	1.3596	0.2488	Fail to Reject H_0	Not Significant
5	1.4270	0.2154	Fail to Reject H_0	Not Significant

The results indicate that the USD/LKR exchange rate does not Granger-cause ASPI at any lag length, as all p-values are greater than the 0.05 significance level.

The Granger causality test results indicate a **unidirectional causal relationship** between ASPI and the USD/LKR exchange rate. The findings suggest that past values of ASPI help predict movements in the USD/LKR exchange rate, particularly in the short run. However, there is no evidence that past values of the exchange rate help predict ASPI movements.

This implies that stock market performance may contain useful information for forecasting exchange rate movements, while exchange rate changes do not significantly influence stock market movements in the short run.

4.6 Cointegration Results

Rank	Trace Statistic	Critical Value (95%)	Decision
0	33.4912	15.4943	Reject Null Hypothesis (Cointegration Exists)
1	0.0007	3.8415	Fail to Reject Null Hypothesis

Table 4.6 : Cointegration Results

The Johansen Cointegration test was conducted to examine whether a long-term equilibrium relationship exists between the USD/LKR exchange rate and ASPI. The test results are presented in *Table 4.6 : Cointegration Results*

The trace statistic value for rank 0 is 33.4912, which is greater than the corresponding critical value at the 95% confidence level of 15.4943. Therefore, the null hypothesis of no cointegration is rejected, indicating the presence of a long-run equilibrium relationship between the variables.

For rank 1, the trace statistic value is 0.0007, which is lower than the critical value of 3.8415. This result suggests that there is only one cointegrating equation between the USD/LKR exchange rate and ASPI.

The existence of cointegration confirms that although short-term fluctuations may occur, the two variables tend to move together over the long run. This finding implies that exchange rate movements have a long-term influence on stock market performance in Sri Lanka.

Since cointegration is present between the variables, the Vector Error Correction Model (VECM) was employed to capture both short-run adjustments and long-run equilibrium dynamics. The VECM results are discussed in the following section.

4.7 Vector Error Correction Model (VECM) Results

The Vector Error Correction Model (VECM) was estimated to analyze both short-run dynamics and long-run equilibrium adjustments between the USD/LKR exchange rate and ASPI. The VECM results consist of short-run coefficient estimates, error correction terms, and long-run cointegration relationships.

4.7.1 Short-Run Dynamics

Table 4.7.1 : Short-Run Dynamics of VECM Results

Dependent Variable: ASPI

Variable	Coefficient	Std. Error	Z-Statistic	P-value	Significance
L1.ASPI	0.0339	0.064	0.527	0.598	Not Significant
L1.USD/LKR	6.6937	18.089	0.370	0.711	Not Significant

Dependent Variable: USD/LKR Exchange Rate

Variable	Coefficient	Std. Error	Z-Statistic	P-value	Significance
L1.ASPI	-0.0006	0.000	-2.820	0.005	Significant
L1.USD/LKR	0.1819	0.062	2.912	0.004	Significant

The short-run dynamics results are presented in *Table 4.7.1 : Short-Run Dynamics of VECM Results*

For the ASPI equation, the lagged ASPI variable and lagged USD/LKR exchange rate were found to be statistically insignificant, as their p-values were greater than the 0.05 significance level. This indicates that short-term changes in both ASPI and exchange rate do not have a significant immediate impact on ASPI movements.

For the USD/LKR exchange rate equation, the lagged ASPI variable recorded a negative and statistically significant coefficient ($p = 0.005$). This suggests that previous changes in ASPI have a short-run influence on exchange rate movements. Additionally, the lagged USD/LKR exchange rate variable was found to be statistically significant ($p = 0.004$), indicating persistence in exchange rate behavior over time.

4.7.2 Error Correction Mechanism

Table 4.7.2: Error Correction (Loading Coefficients)

Equation	Error Correction Term (ec1)	Std. Error	Z-Statistic	P-value	Interpretation
ASPI	-0.0028	0.001	-3.591	0.000	Significant Long-Run Adjustment
USD/LKR	-0.0000003418	0.00000266	-0.128	0.898	Not Significant

The error correction terms measure the speed at which variables adjust towards long-run equilibrium following short-term deviations.

The error correction coefficient for the ASPI equation was found to be negative and statistically significant (coefficient = -0.0028, $p < 0.05$). This confirms the presence of long-run equilibrium adjustment and indicates that deviations from long-run equilibrium are gradually corrected over time. The negative sign suggests that ASPI adjusts towards restoring equilibrium when disequilibrium occurs.

In contrast, the error correction term for the USD/LKR exchange rate was statistically insignificant ($p = 0.898$). This suggests that exchange rate adjustments towards long-run equilibrium are relatively weak compared to ASPI adjustments.

4.7.3 Long-Run Cointegration Relationship

Table 4.7.3: Long-Run Cointegration Relationship

Variable	Coefficient	Std. Error	Z-Statistic	P-value
ASPI	1.0000	—	—	0.000
USD/LKR	-104.1785	13.904	-7.493	0.000

The long-run cointegration relationship between ASPI and USD/LKR exchange rate is presented in *Table 4.7.3: Long-Run Cointegration Relationship*

The results indicate that the USD/LKR exchange rate has a statistically significant negative relationship with ASPI in the long run. The estimated coefficient (-104.1785) suggests that an increase in the exchange rate is associated with a decrease in ASPI over time.

This finding implies that currency depreciation may negatively influence stock market performance in Sri Lanka in the long run.

The VECM results confirm the existence of both short-run and long-run relationships between the USD/LKR exchange rate and ASPI. While short-run impacts on ASPI were found to be limited, the significant error correction mechanism indicates that ASPI adjusts to maintain long-term equilibrium with exchange rate movements.

These findings support the importance of exchange rate fluctuations in influencing stock market performance and justify the use of cointegration-based econometric modelling techniques.

4.8 Model Performance Comparison Overview

To identify the most appropriate model for forecasting the All Share Price Index (ASPI), a comprehensive performance comparison was conducted across all developed models. The comparison includes traditional linear regression models, dynamic regression models with lagged variables, and machine learning techniques such as Random Forest, Ridge Regression, and Lasso Regression.

Each model was evaluated using multiple accuracy metrics, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Normalized Mean Squared Error (NMSE), the coefficient of determination (R^2), and Mean Absolute Percentage Error (MAPE). These evaluation criteria were selected to ensure a balanced assessment of prediction accuracy, error magnitude, explanatory power, and overall model reliability.

Lower values of MSE, RMSE, MAE, NMSE, and MAPE indicate better predictive performance, as they represent smaller deviations between actual and predicted ASPI values. Conversely, a higher R^2 value suggests that the model explains a larger proportion of the variation in the dependent variable.

All models were trained and tested using the same dataset and consistent preprocessing procedures to ensure fairness in comparison. The following table summarizes the performance of each model and provides a clear basis for selecting the most accurate and reliable forecasting model.

4.8.1 Model Performance Comparison Table

Model	MSE	RMSE	MAE	NMSE	R²	MAPE (%)
Simple Linear Regression Model	3158869.62	1777.32	1267.86	0.4635	0.5354	8.3466
Multiple Linear Regression with One Lag of USD	2751051.68	1658.63	1238.90	0.4087	0.591	8.0522
Multiple Linear Regression with Two Lags of USD	2423832.21	1556.87	1200.71	0.3642	0.636	7.7438
Multiple Linear Regression with Three Lags of USD	2135809.47	1461.44	1163.62	0.3240	0.676	7.4548
Multiple Linear Regression with Four Lags of USD	1937795.57	1392.05	1126.29	0.2969	0.703	7.1788
Multiple Linear Regression with Five Lags of USD	1787848.22	1337.10	1095.37	0.2766	0.723	6.9493
Multiple Linear Regression with Five Lags of USD and One Lag of ASPI	28507.30	168.84	118.75	0.0044	0.996	0.7261
Multiple Linear Regression	28481.82	168.77	118.31	0.0044	0.996	0.7235

with Five Lags of USD and Two Lags of ASPI						
Multiple Linear Regression with USD Lag 1 and ASPI Lag 1 (Without Actual USD)	29245.28	171.01	121.01	0.0043	0.995	0.7451
Ridge Regression (Test Data)	20237.13	142.26	112.68	0.0419	0.958	0.553
Lasso Regression (Test Data)	17707.49	133.07	104.44	0.0367	0.963	0.513
Random Forest Regression	4319984.47	2078.45	1959.02	8.9587	-7.95	9.5714
Model	MSE	RMSE	MAE	NMSE	R²	MAPE (%)

Table 4.8.1 : Model Performance Comparison Results

Table 4.8.1 : Model Performance Comparison Results presents a comprehensive comparison of all developed regression and machine learning models based on multiple evaluation metrics, including MSE, RMSE, MAE, NMSE, R², and MAPE.

The results clearly demonstrate that the prediction accuracy improves progressively as lagged variables are introduced into the regression models. The Simple Linear Regression model shows relatively high error values (MSE = 3,158,869.62; RMSE = 1777.32; MAPE = 8.35%) and a comparatively low R² value (0.5354), indicating moderate explanatory power and limited forecasting accuracy.

When additional lags of the USD/LKR exchange rate were included, the performance gradually improved. The Multiple Linear Regression model with five lags of USD achieved lower error values (MSE = 1,787,848.22; RMSE = 1337.10; MAPE = 6.95%) and a higher R² value (0.723), demonstrating that incorporating lagged exchange rate information enhances predictive capability.

However, a significant improvement is observed when lagged ASPI values are included in the model. The Multiple Linear Regression model with five lags of USD and one lag of ASPI dramatically reduced the prediction error ($MSE \approx 28,507$; $RMSE \approx 168.84$; $MAPE \approx 0.73\%$) and achieved an exceptionally high R^2 value of 0.996. This indicates that the inclusion of ASPI lag variables captures the strong autoregressive nature of the stock market index. Similar performance is observed in the model with two ASPI lags, confirming the dominance of past ASPI values in predicting current market levels.

Among the machine learning models, Ridge Regression and Lasso Regression (evaluated on test data) demonstrate superior forecasting performance compared to traditional regression models without ASPI lags. The Lasso Regression model achieved the lowest error values ($MSE = 17,707.49$; $RMSE = 133.07$; $MAE = 104.44$) and the highest R^2 value (0.963) with a very low MAPE of 0.513%. This suggests excellent generalization ability and high predictive accuracy.

In contrast, the Random Forest model produced significantly higher error values ($MSE = 4,319,984.47$; $RMSE = 2078.45$) and a negative R^2 value (-7.95), indicating poor predictive performance on the test data. This suggests that the Random Forest model may have overfitted the training data and failed to generalize effectively.

Overall, the findings confirm that models incorporating lagged ASPI variables substantially outperform models relying solely on exchange rate information. Furthermore, among all evaluated models, **the Lasso Regression model provides the best balance between accuracy and generalization performance, making it the most suitable model for forecasting ASPI in this study.**

5. Conclusion

This study examined the relationship between the USD–LKR exchange rate and the All Share Price Index (ASPI) in Sri Lanka using both econometric and machine learning techniques. The primary objective was to develop a reliable statistical model capable of accurately forecasting ASPI movements during a recent economic transition period.

The descriptive analysis revealed noticeable volatility and dynamic behavior in both ASPI and the exchange rate series. Stationarity testing using the Augmented Dickey-Fuller (ADF) test confirmed that both variables were non-stationary in their level form but became stationary after first differencing, indicating that the series are integrated of order one, $I(1)$.

The Johansen cointegration test identified the presence of a long-run equilibrium relationship between the USD/LKR exchange rate and ASPI. This confirms that although short-term fluctuations occur, the two variables move together over time. The Vector Error Correction Model (VECM) results further revealed a significant error correction mechanism for the ASPI equation, indicating that deviations from long-run equilibrium are gradually corrected. This highlights the existence of long-term interdependence between exchange rate movements and stock market performance.

However, short-run dynamics showed limited direct impact of exchange rate changes on immediate ASPI movements. Instead, the results consistently demonstrated strong autoregressive behavior in ASPI. The inclusion of lagged ASPI variables significantly improved model performance, indicating that past stock market values are the dominant determinant of current market levels.

The model comparison analysis revealed substantial improvements in forecasting accuracy when lagged ASPI variables were incorporated. Traditional regression models relying solely on exchange rate information exhibited moderate explanatory power. In contrast, dynamic regression models with ASPI lag terms achieved exceptionally high R^2 values and significantly lower prediction errors.

Among the machine learning approaches, the Lasso Regression model demonstrated the best overall performance, achieving the lowest MSE, RMSE, and MAPE values, along with a high R^2 value. This indicates strong predictive accuracy and good generalization capability. The Ridge Regression model also performed well, while the Random Forest model exhibited comparatively poor performance on test data, suggesting potential overfitting.

Overall, the findings of this study suggest that while exchange rate fluctuations have a statistically significant long-run relationship with stock market performance, short-term ASPI movements are primarily driven by internal market momentum and historical price persistence. The results confirm that incorporating autoregressive components is essential for accurate stock market forecasting.

This study contributes to the existing literature by combining cointegration-based econometric modelling with regularized machine learning techniques using recent daily data from Sri Lanka. The findings provide practical implications for investors, financial analysts, and policymakers by offering a stable and reliable forecasting framework for short-term ASPI prediction.

In conclusion, penalized regression methods, particularly Lasso Regression, provide an effective and robust approach for forecasting stock market indices in emerging economies such as Sri Lanka, especially when strong autoregressive characteristics are present.

6.The Reference Materials

6.2 References

- Anusha, P., Dhushanthan, B., & Vinayagathan, T. (2022). The relationship between exchange rate and stock market performance: Empirical evidence from Sri Lanka. *Business and Economic Research*, 135-154.
- Balogobei, S. &. (2022). Impact of macroeconomic variables on stock market performance: Evidence from Sri Lanka. *Wayamba Journal of Management*, 28–45.
- Balogobei, S. (2017). Macroeconomic variables and stock market returns in Sri Lanka. . *Asian Journal of Finance and Accounting*, 118–132.
- Balcilar, M. B. (2020). The foreign exchange and stock market nexus: New international evidence. . *International Review of Economics & Finance*, 240–266.
- Gokmenoglu, K. E. (2021). Exchange rates and stock markets in emerging economies: New evidence using the quantile-on-quantile approach. . *Quantitative Finance and Economics*, 94-110.
- Gooneratne, D. W. (2022). The effect of foreign exchange rates on the stock market performance in Sri Lanka. *Journal of Management Matters*, 39–62.
- Jayasundara, J. M. (2019). Impact of macroeconomic variables on stock market performances: Evidence from Sri Lanka. . *Sri Lankan Journal of Business Economics*, 1–18.
- Kengatharan, L. &. (2021). Macroeconomic variables and stock market performance in Sri Lanka: An ARDL bound testing approach. *South Asian Journal of Business Insights*, 43–58.
- Menike, L. M. (2010). The effect of macroeconomic variables on stock prices in emerging Sri Lankan stock market. *Sabaragamuwa University Journal*, 50–67.

Sujenthini, S. &. (2022). A study on effects of exchange rate fluctuation on stock returns: Evidence from Colombo Stock Exchange. *EPRA International Journal of Economics, Business and Management Studies*, 88-97.

6.2 Appendices

Google Colab Link : <https://colab.research.google.com/drive/1uZAJuUUIYDE87OVVwQv6-ixkY5FaBbM1?usp=sharing>

7.Individual Contribution

	Student Number	Task Completed
01	PS/2021/167	✓ Collected ASPI and USD/LKR exchange rate data from relevant sources. ✓ Conducted literature review and gathered references.. ✓ Assisted in correlation analysis and graphical presentation.
02	PS/2021/168	✓ Implemented cointegration analysis and VECM model. ✓ Assisted in econometric model development. ✓ Contributed to the methodology section.
03	PS/2021/171	✓ Developed regression models including OLS, Lasso, and Ridge. ✓ Calculated model performance metrics. ✓ Contributed to the machine learning section.
04	PS/2021/057	✓ Performed data cleaning and preprocessing ✓ Applied differencing techniques to achieve stationarity. ✓ Assisted in interpreting statistical test results.
05	PS/2021/135	✓ Performed model comparison and evaluation. ✓ Prepared result tables and summaries. ✓ Assisted in the Results and Discussion section.
06	PS/2021/176	✓ Conducted Exploratory Data Analysis (EDA). ✓ Created time series plots and visualizations. ✓ Assisted in descriptive statistical analysis.
07	PS/2021/112	✓ Performed stationarity testing using ADF tests. ✓ Applied differencing techniques to achieve stationarity. ✓ Assisted in interpreting statistical test results.
08	PS/2021/033	✓ Formatted and edited the final report. ✓ Prepared the Table of Contents and lists of tables and figures. ✓ Compiled and finalized the report for submission.